Earth and Planetary Science Letters, 94 (1989) 143–150
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[5]

An Upper Cretaceous paleomagnetic pole for stable Europe from Aix-en-Provence (France)

Michel Westphal [1] and Jean Pierre Durand [2]

[1] *Institut de Physique du Globe, 5 rue R. Descartes, F-67084 Strasbourg Cedex (France)*
[2] *Laboratoire de Géologie, Université d'Aix Marseille II, Faculté des Sciences de Luminy, 70 route L. Lachamp, F-13288 Marseille Cedex 09 (France)*

Received March 7, 1989; revised version accepted May 3, 1989

A paleomagnetic study of continental deposits (limestones, sandstones) from the Upper Cretaceous basin of Aix-en-Provence (Campanian to Danian) (43.5° N, 5.5° E) gives a mean paleomagnetic direction of: $D = 9.5°$, $I = 46.8°$, $N = 10$, $k = 33$, $\alpha_{95} = 8.3°$ and a mean pole: 73° N, 156° E, $N = 10$, $K = 24$, $A_{95} = 10°$. The fold test is positive, as is the reversal test. This pole is not significantly different from other Cretaceous or lower Tertiary poles from stable Europe.

1. Introduction

Paleomagnetism is widely used for detecting tectonic movements in mobile belts such as the Alpine-Mediterranean area (see for instance [1,2]). But the interpretations are always hampered by the lack of good reference poles for the stable parts of the plates: extra-alpine Europe and Africa. We are presently in a situation where the polar wander curves of mobile zones such as Apulia are better known than those for stable Europe or stable Africa.

For the Cretaceous period only a few reference poles are available for Europe: one from Berrias stratotype section in southern France [3], two from Germany [4,5] in the Munster basin, two sections in the Isle of Wight [6] and a recent result from Aubeterre in southwestern France [7]. For the Paleocene several results from the lavas and the dykes from the British Tertiary igneous province are available (Mull, Eigg, Muck, Skye islands and Antrim), [8–16]. Studies in stable Europe are difficult because of the lack of good outcrops with fold tests and the usual weakness of the magnetizations in sediments.

We present here the results of a study of Late Cretaceous continental deposits from the region of Aix-en-Provence in France. These series were studied mainly for magnetostratigraphy. But several section gave good paleomagnetic directions that allowed us to calculate a reliable paleomagnetic pole.

2. Geology and sampling

The late Cretaceous basin of Aix-en-Provence is a wide east–west syncline of about 400 km^2. The deposits are mainly lacustrine and range from Santonian to Eocene. This basin is well known for its dinosaurs egg shells. The lack of marine fossils led the geologists to define a local stratigraphy based on lithologic properties [17–22]. The magnetostratigraphic study of the basin allowed us to correlate some of the local stages with marine stages (Westphal and Durand, in preparation).

The oldest stage, the Valdonnian, of normal polarity, is correlated with the Santonian. The following ones: Fuvelian and Begudian show a reversed period followed by a long normal one and are correlated with the Campanian. These stages are mainly formed of fluviatile and lacustrine limestones and marls. Brown coal seams are frequent, mainly in the Fuvelian where they are still exploited. The Maastrichtian, where several reversals are found, is represented by more detrital deposits: sandstones, limestones, redbeds, called Rognacian. The Cretaceous–Tertiary boundary lies probably near a huge conglomerate

0012-821X/89/$03.50 © 1989 Elsevier Science Publishers B.V.

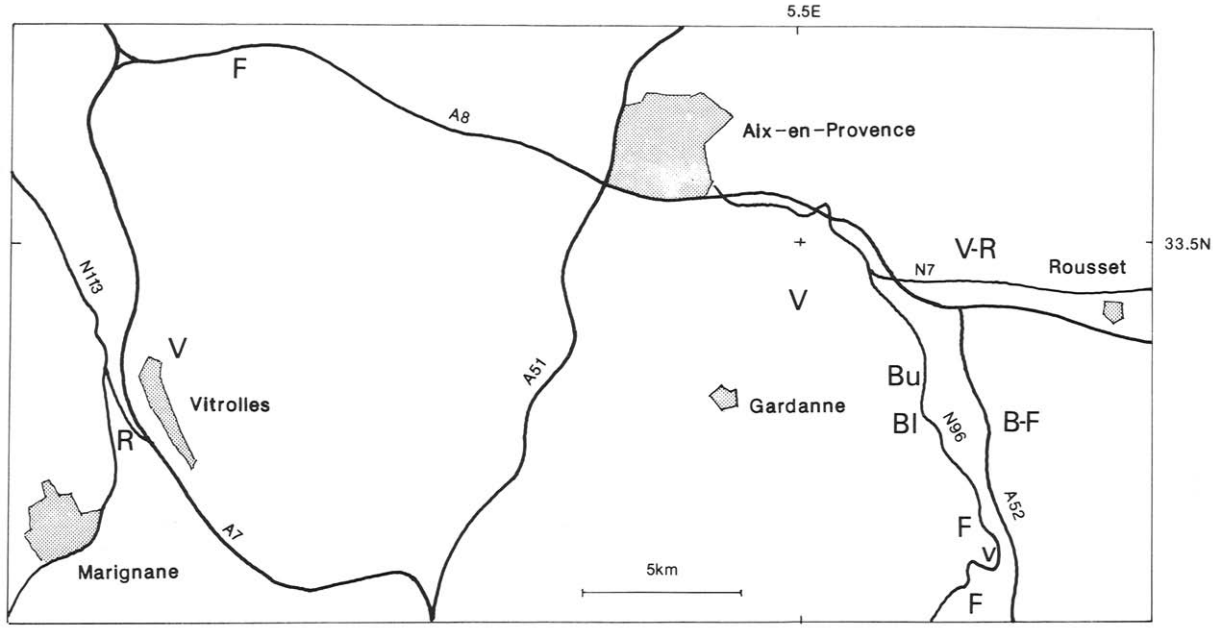


Fig. 1. Outline of the region sampled, with main roads, highways and towns. *V* shows the Vitrollian sites (5 and 14); *V-R*, the Vitrollian–Rognacian section: *R*, the limestone of Rognac section; *Bu*, *Bl* the two Begudian sections; *B-F*, the Begudian–Fuvelian section; *F*, the Fuvelian sections (1, 4 and 32); and *v* the Valdonnian section.

called Poudingue de la Galante, separating the Rognacian from the Vitrollian. This last stage, represented by red or pink sandstones, marls, limestones is thought to correspond to the Danian.

Sampling was done in several sections either by handsampling or with a portable drill. We shall describe here only the sections from which our paleomagnetic results are obtained (Fig. 1).

A single section of the Valdonnian was sampled along the road N96 (site *v* in Fig. 1). The section sampled is 50 m thick, but it did not give very good directions. We took samples from the Fuvelian in three different outcrops (sites *F*): just above the Valdonnian, along the N96, in the big quarry of Valdonne and near Coudoux in the northwest of the basin. We sampled the transition with the Begudian (site *B-F*) along the highway A52. The Begudian itself was sampled along the road N96 between the Jas de Bassas and La Barque. This section is cut in two subsections (sites *Bl* and *Bu*). The different levels of the Rognacian were sampled in different places, but we get reliable results only in one section in the limestone between Rognac and Vitrolles (site *R*). The redbeds of the Vitrollian were sampled near Vitrolles, northeast of Gardanne and near Rousset (sites *V* and *V-R*).

All the samples were cut in standard 25 mm cores and the measurements were done with a modified Digico spinner magnetometer. The noise level of our apparatus was always less than 0.04 mA/m. All the samples with a magnetization higher than 0.5 mA/m were demagnetized by step heatings or in some cases by alternating field. Rock magnetic properties were controlled by IRM acquisition curves and the corresponding thermal decay curves (Fig. 2). For several samples the low field susceptibility was checked after each heating step. The low field anisotropy of susceptibility was also measured, but it is very weak and has no significance.

3. Results

The rock magnetic properties show three main types of samples. First, samples where the main magnetic mineral is goethite. The IRM acquisition curves show a constant increase in magnetization up to 1.5 T, the highest field that we can obtain, with a sharp decrease of the IRM magnetization

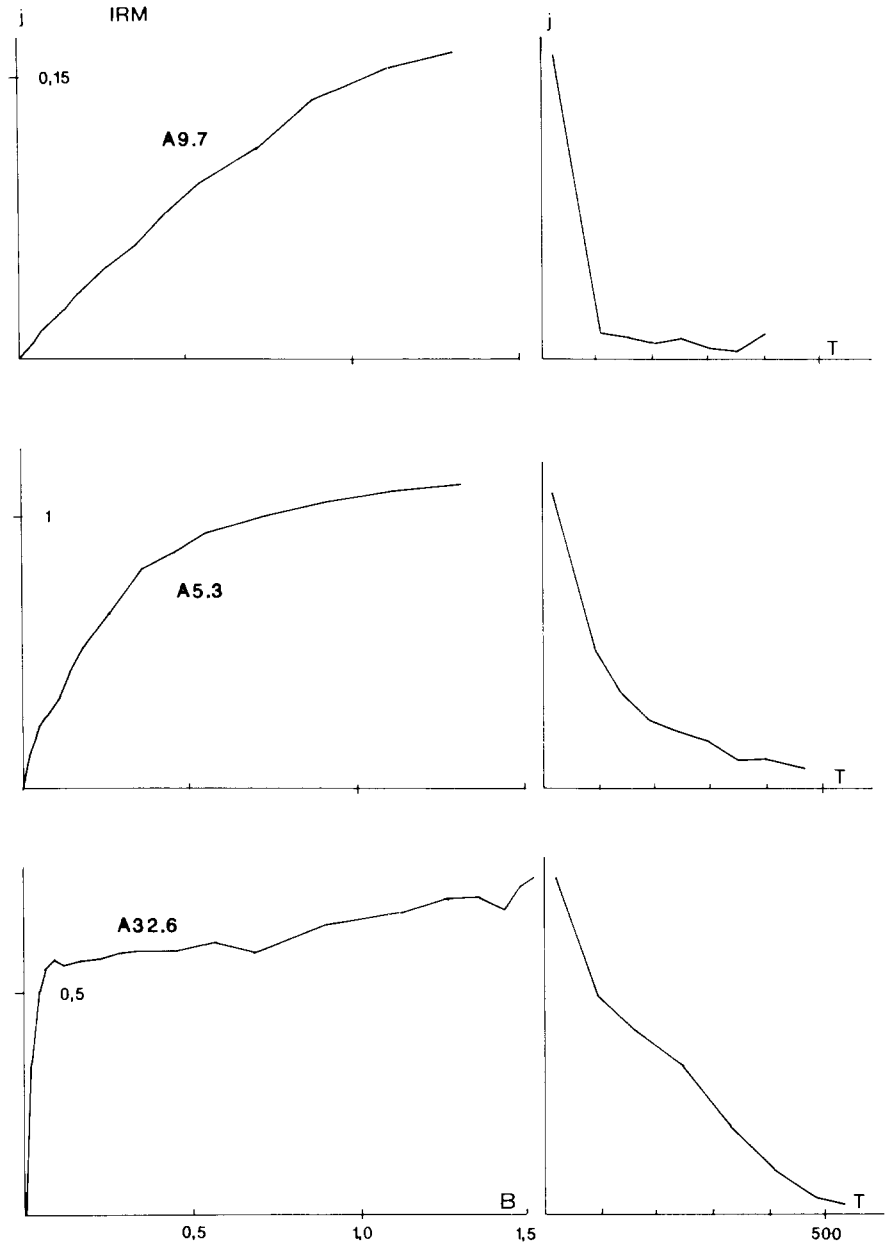


Fig. 2. IRM acquisition curves (left) with the applied field and the corresponding thermal decay with temperature (right). Intensities (J) are in A/m, fields (B) in Teslas and temperatures (T) in °C. A9.7: a Begudian limestone with mainly goethite; A5.3: a Vitrollian limestone with haematite; A32.6: a Fuvelian limestone with probably sulphides.

after heating at 100 °C (Fig. 2, sample A9.7). Such samples are found mainly in the yellow or beige limestones from the Valdonnian and the Begudian. The NRM of these samples are often very weak and also drops sharply at 100 °C below or close to the noise level of our Digico. Therefore they did not give reliable results.

A second family shows a rather quick saturation of the IRM with fields of about 0.1 or 0.2 T (Fig. 2, sample A32.6). The unblocking tempera-

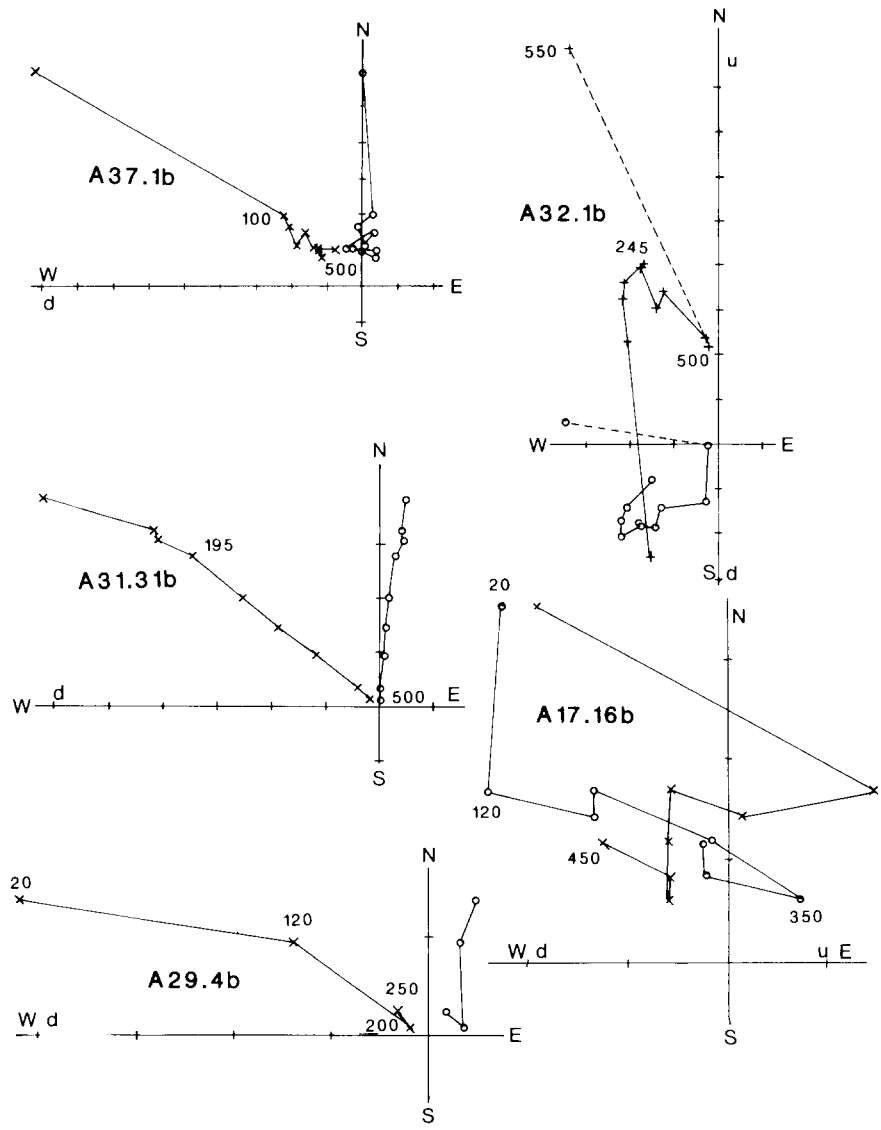


Fig. 3. Thermal demagnetization curves for several samples. ○ = north–south horizontal plane, × = north–south vertical plane, + = east–west vertical plane. N, E, S, W = north, east, south and west directions; d, u = down and up directions. Scales are 0.1 mA/m except for A31.31b where it is 1 mA/m. A37.1b, A31.31b, A32.1b are examples of samples used for this paper (Rognacian, Begudian and Fuvelian limestones). The last thermal step of sample A32.1b shows a strong increase in magnetization due to the mineralogical changes in the sample (probably due to sulphides). A29.4b, A17.16b are examples of samples *not used* for this paper (a Valdonnian limestone and a Fuvelian sandstone), the first one has a too low blocking temperature and the second one did not show a clear magnetization direction.

ture of the IRM is above 500 °C, but strong modifications of the magnetic properties occur above 400 °C as shown by the ten- to hundred-fold increase of the saturation IRM given after heating at this temperature. The NRM unblocking temperature is about 350–400 °C. Such samples are mainly found in the Fuvelian. The magnetic carriers are probably sulphides which is consistent with the presence of brown coal showing that the medium was reducing in these levels. Usually clear characteristic magnetization could be calculated for these samples.

The third group is given by the red samples. The IRM does not saturate, but the intensity of

TABLE 1

Site mean results

Site	N	D (°)	I (°)	D' (°)	I' (°)	k	α_{95} (°)	Poles	
								°N	°E
Vitrollian									
site 5	8	359	49	359	49	9	18	77	189
site 14	7	55	61	39	45	36	10	54	111
Rognacian/Vitrollian	13	22	61	4	52	7	16	79	168
Rognacian limestone	9	342	44	342	44	13	14	67	231
Begudian u.	11	22	53	23	40	11	14	62	136
Begudian l.	9	9	46	4	44	15	13	72	174
Begudian/Fuvelian	15	354	54	345	45	16	10	69	226
Fuvelian									
site 4	9	207	−38	193	−50	67	6	74	142
site 1	7	200	−4	202	−32	66	7	58	143
site 32	8	212	−60	202	−54	52	8	71	116
Means	10	14.8	48.9			14.6	12.6		
	10			9.5	46.8	33.2	8.3		
Normal sites	7			5	47	30	11		
Reversed sites	3			199	−46	44	19		
Mean pole	10					23.8	10	73	156

N = number of samples used in the statistics; D, I = declination and inclination in situ; D', I' = declination and inclination in stratigraphical coordinates; k, α_{95} = Fisher statistical parameters.

magnetization is higher than for the first group and the thermal decay of the magnetization is much slower (Fig. 2, sample A5.3). The unblocking temperature of the IRM magnetization is above 500 °C as for the NRM. Most of the upper Rognacian and Vitrollian samples belonged to this group. The magnetization is carried mainly by haematite in these samples. Usually a good characteristic magnetization could be found, although some remagnetizations exist. Combinations of the

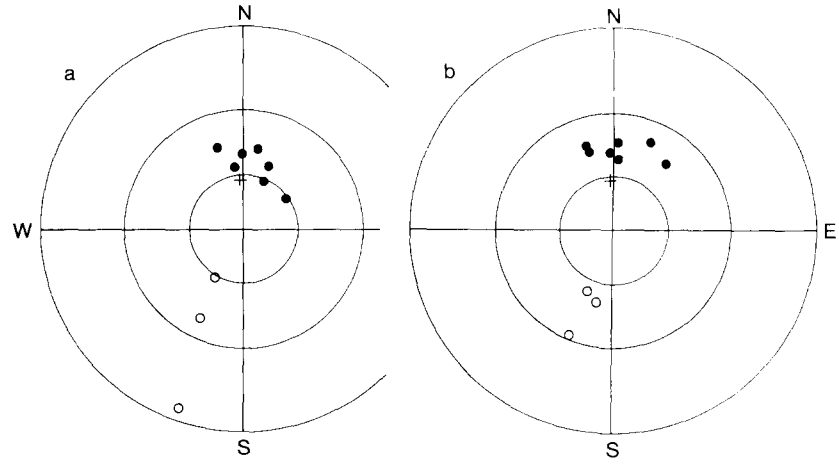


Fig. 4. Site mean directions before (a) and after tectonic corrections (b). Stereographic projection; filled circles = positive inclinations; open circles = negative inclinations; crosses = present magnetic field.

two first types are also found mainly in the Fuvelian.

Among all the samples taken, one third of the NRM were too weakly magnetized or showed a too quick thermal decay at 100 °C to allow a further thermal demagnetization. For the purpose of this paper we selected from the other samples those that showed a fairly good demagnetization curve allowing us to calculate a characteristic magnetic vector (Fig. 3). We could get reliable directions from 4 sections in the Fuvelian, 2 from the Begudian, 1 from the Rognacian and 3 from the Vitrollian. In other sections sampled the directions were too scattered for our purpose ($k < 10$) or we could only obtain 2 or 3 reliable directions per site. The results retained are shown in Table 1.

Most of the sections have a slight northward dip of about 10°. Luckily a few ones have a different dip allowing us to make a fold test. Taking all sites together the ratio of k_2 to k_1 is 2.23 and is significant at the 95% level, but not at the 99% level [23]. The fold test is positive. It is mainly due to the three reversed sections from the Fuvelian (Fig. 4). The normal and reversed direction are nearly opposite, the difference in declination is about 14° and is 1° in inclination. A common present field overprint on both normal and reversed directions should have given a difference between inclinations, but not between declinations. Thus we consider that this difference (11°) is not significant and that the reversal test is also positive and indicates that the magnetic cleaning is good.

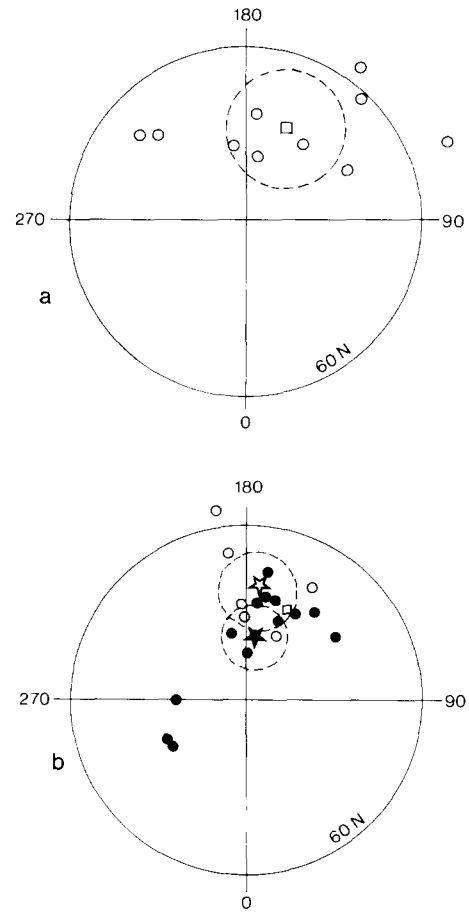


Fig. 5. (a) VGPs of site means from Aix-en-Provence. The square is the overall mean pole. The confidence circle around the mean pole is shown. (b) Paleomagnetic poles from stable Europe. Open symbols = Cretaceous poles, the square is the pole from Aix-en-Provence, the star is the overall mean of all Cretaceous poles. Filled circles = Paleocene poles from the British Tertiary igneous province. The black star is the corresponding overall mean. Confidence circles around mean poles are shown.

4. Comparison with other results

A first study in the Aix-en-Provence basin was already done in 1974 by Biquand [24] on some Vitrollian limestones. He demagnetized his samples mainly by a few alternating field or by a few heating steps. His result (pole: 70 °N, 156 °E on one site and 12 samples) is similar to ours. This result is now superseded by ours. Several studies were also done for magnetostratigraphy in this basin by Galbrun, Hansen, Krumsiek, but they did not yet publish their mean directions.

Fig. 5 shows the poles obtained for Cretaceous formations in Europe and from some Paleocene volcanics from the British Tertiary igneous province. The differences between all these are small (Table 2).

The reliability of the different Cretaceous results is variable. All the results are obtained after magnetic cleaning, but most are based on single sections, without any fold test, whereas one of the studies in the Munster basin [4] and the present one, are based on several sections and the stability of the characteristic magnetization is controlled by a fold test.

There is no obvious age trend between Cretaceous poles, although there is a large difference in

TABLE 2

European Paleocene and Cretaceous poles

Name	N	Pole		α_{95} (°)	Reference
		°N	°E		
Paleocene, British Tertiary igneous province					
Mull island					
lavas	78	72	168	3	[15]
ring dyke *	6	75	298	10	[8]
normal dykes *	39	75	301	3	[9]
reversed dykes	115	73	174	3	[9]
Fishnish dykes	14	71	142	9	[11]
Skye island					
lavas	90	72	165	3	[16]
Vaternish normal dykes *	18	78	270	10	[15]
Vaternish reversed dykes	68	73	152	5	[15]
Arran island	413	82	180	2	[12]
Eigg island					
lavas	34	73	174	4	[11]
dykes	42	75	158	4	[11]
Muck island					
lavas	21	78	190	3	[11]
dykes	30	68	171	5	[11]
Ireland, Antrim basin	19	71	126	24	[13]
Mean Paleocene poles	14	79	174	6	$K = 41$
Selected poles	11	74	162	4	$K = 166$
Cretaceous					
Munster basin (Turonian)	1	68	149	5	[5]
Munster basin (Cenomanian–Campanian)	11	76	181	4	[4]
Aubeterre (Campanian)	1	78	155	3	[7]
Isle of Wight (Aptian–Barremian)					
Sheperd ch. *	1	65	187	9	[8]
Yaverland *	1	58	189	9	[8]
Ardèche (Berriasian)					
Berrias	1	74	183	3	[3]
Aix-en-Provence (Campanian–Danian)	10	73	156	10	this paper
Mean Cretaceous pole					
all poles	7	71	174	7	$K = 78$
without Isle of Wight	5	74	164	6	$K = 194$

N = number of sites or of sections, α_{95} = confidence cone of the results. * Indicate possible outliers (see text).

age between them. The two poles from the Isle of Wight are a little away from other Cretaceous poles. Kerth and Hailwood [6] suggest that the difference is due to a depositional error or to compaction effects.

Three poles from the British Tertiary igneous province show also a discrepancy from the other eleven poles from the same province. It is the Vaternish normal dykes from Skye island, the dykes cutting the Loch Ba ring dyke and the normal dykes from Mull island. Looking back in the publications we could not find a clear explanation for these differences.

We have then two possibilities. We can consider that the differences between the two poles form the Isle of Wight and other Cretaceous poles on one hand and within the British Tertiary province on the other hand are not significant. In this case there is a small trend of the mean poles between the Cretaceous and the Paleocene. The second possibility is to consider that these differences are significant and that we must reject

these outliers. In this case the difference between Cretaceous and Paleocene poles of stable Europe is insignificant. But it is too easy to reject directions just because they are different from others and we should find a better explanation for them. We prefer to take all the results until good arguments are found.

Acknowledgements

The authors are grateful to B. Galbrun who communicated us the pole from Aubeterre and to a second reviewer for helpful comments. The field and laboratory works were supported by CNRS and INSU grants. This is a contribution of CNRS UA 323.

References

1 M. Westphal, M. Bazhenov, J.P. Lauer, D.M. Pechersky and J.C. Sibuet, Paleomagnetic implications on the evolution of the Tethys belt from the Atlantic ocean to the Pamirs since the Triassic, Tectonophysics 123, 37–82, 1986.
2 W. Lowrie, Paleomagnetism and the adriatic promontory: a reappraisal, Tectonics 5(5), 797–807, 1986.
3 B. Galbrun, Magnetostratigraphy of the Berriasian stratotype section (Berrias, France), Earth Planet. Sci. Lett. 74, 130–136, 1985.
4 F. Heller and J.E.T. Channell, Paleomagnetism of Upper Cretaceous limestones from the Münster basin, Germany, J. Geophys. 46, 413–427, 1979.
5 M. Kerth, A paleomagnetic study of Turonian carbonates from the southeastern Münsterland area, NW Germany, J. Geophys. 57, 118–124, 1985.
6 M. Kerth and E.A. Hailwood, Magnetostratigraphy of the Lower Cretaceous Vectis Formation (Wealden Group) on the Isle of Wight, southern England, J. Geol. Soc. London 145, 351–360, 1988.
7 S. Clauser, Etudes stratigraphiques du Campanien et du Maastrichtien de l'Europe occidentale, Côte basque, Charentes (France), Limbourg (Pays Bas), 245 pp., Thesis, Paris, 1988.
8 J.M. Ade-Hall, The palaeomagnetism of some basic dykes cutting the Loch Ba felsite ring dyke, isle of Mull, Scotland. Geophys. J. R. Astron. Soc. 36, 267–271, 1974.
9 J.M. Ade-Hall, P. Dagley, R.L. Wilson, A. Evans, A. Riding, P.J. Smith, R. Skelhorne and T. Sloan, A palaeomagnetic study of the Mull regional dyke swarm, Geophys. J. R. Astron. Soc. 27, 517–545, 1972.
10 P. Dagley and A.E. Mussett, Paleomagnetism of the Fishnish dykes, Mull, Scotland. Geophys. J. R. Astron. Soc. 53, 553–558, 1978.
11 P. Dagley and A.E. Mussett, Palaeomagnetism and radiometric dating of the British tertairy igneous province: Muck and Eigg, Geophys. J. R. Astron. Soc. 85, 221–242, 1986.
12 P. Dagley, A.E. Mussett, R.L. Wilson and J.M. Hall, The British Tertiary igneous province: palaeomagnetism of the Arran dykes, Geophys. J. R. Astron. Soc. 54, 75–91, 1978.
13 J.M. Hall, R.L. Wilson and P. Dagley, A paleomagnetic study of the Mull lava succession, Geophys. J. R. Astron. Soc. 49, 499–514, 1977.
14 R. Løvlie, A. Gidskehaug and K.M. Storetvedt, On the magnetization history of the northern irish basalts, Geophys. J. R. Astron. Soc. 27, 487–498, 1972.
15 R.L. Wilson, J.M. Ade-Hall, R.R. Skelhorn, J.M. Speight and P. Dagley, The British Tertiary igneous province: palaeomagnetism of the Vaternish dyke swarm on north Skye, Scotland, Geophys. J. R. Astron. Soc. 37, 23–30, 1974.
16 R.L. Wilson, P. Dagley and J.M. Ade-Hall, Palaeomagnetism of the British Tertiary igneous province: the Skye lavas, Geophys. J. R. Astron. Soc. 28, 285–293, 1972.
17 J.F. Babinot and J.P. Durand, Le Valdonnien. Les étages français et leurs stratotypes, Mém. B.R.G.M. 109, 171–174, 1980.
18 J.F. Babinot and J.P. Durand, Le Fuvélien. Les étages français et leurs stratotypes, Mém. B.R.G.M. 109, 175–178, 1980.
19 J.F. Babinot and J.P. Durand, Le Bégudien. Les étages français et leurs stratotypes, Mém. B.R.G.M. 109, 179–183, 1980.
20 J.F. Babinot and J.P. Durand, Le Rognacien. Les étages français et leurs stratotypes, Mém. B.R.G.M. 109, 184–188, 1980.
21 J.F. Babinot and J.P. Durand, Le Vitrollien. Les étages français et leurs stratotypes, Mém. B.R.G.M. 109, 189–192, 1980.
22 J.P. Durand, Les sédiments fuvéliens du synclinal de l'Arc (Provence), Ind. Minér., Suppl., Le gisement de charbon du bassin de l'Arc, pp. 13–25, 1980.
23 M.W. McElhinny, Statistical significance of the fold test in palaeomagnetism, Geophys J. R. Astron. Soc. 8, 338–340, 1964.
24 D. Biquand, L'aimantation rémanente naturelle des roches sédimentaires, ses propriétés, sa nature. Conditions et limites d'une utilisation paléomagnétique, 3 vols., 191 pp., Thesis, Paris, 1974.